

One Hot Encoding

What is one hot encoding?

One hot encoding is a process of **converting categorical data variables** so they can be provided to machine learning algorithms to improve predictions.

One hot encoding is one method of converting data to prepare it for an algorithm and get a better prediction.

With one-hot, we convert each categorical value into a new categorical column and assign a binary value of 1 or 0 to those columns.

Each integer value is represented as a binary vector. All the values are zero, and the index is marked with a 1.

Take a look at this chart for a better understanding:

Type		Type	AA_Onehot	AB_Onehot	CD_Onehot
AA	Onehot encoding →	AA	1	0	0
AB		AB	0	1	0
CD		CD	0	0	1
AA		AA	0	0	0

One hot encoding makes our training data more useful and expressive, and it can be rescaled easily. By using numeric values, we more easily determine a probability for our values. In particular, one hot encoding is used for our output values, since it provides more nuanced predictions than single labels.

Practical

Load dataset & library

```
import pandas as pd
import numpy as np
import seaborn as sns
df=sns.load_dataset('tips')
df
```

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One Hot Encoding

now convert dataset into dummy dataframe for encoding

```
dummy_df = pd.get_dummies(df)
dummy_df
```

you can visualize that there is more converted value or variable which machine can understand

```
pd.get_dummies(  
    data,  
    prefix=None,  
    prefix_sep='_',  
    dummy_na: 'bool' = False,  
    columns=None,  
    sparse: 'bool' = False,  
    drop_first: 'bool' = False,  
    dtype: 'Dtype | None' = None,  
)
```

There is also some columns need not to required for remove this columns

```
pd.get_dummies(df, drop_first=True)
```

One-Hot Encoding with Scikit-learn

Import necessary library

```
from sklearn.preprocessing import OneHotEncoder
```

create object for onehotencoding

sparse= is a matrix we need array for 2dimention so that

sparse =False

One Hot Encoding

use fit_transform method to convert in one hot encoding

we need only categorical data to transform

```
oh_enco_arr=oh_enco.fit_transform(df[['sex','smoker','day','time']])  
oh_enco_arr
```

now we have 2d array

need to convert them into dataframe

we have dummy dataframe keys

```
oh_enco_df=pd.DataFrame(oh_enco_arr,columns=['sex_Male', 'sex_Female', 'smoker_Yes',  
                                             'smoker_No', 'day_Thur', 'day_Fri', 'day_Sat', 'day_Sun',  
                                             'time_Lunch','time_Dinner'])  
oh_enco_df
```